

Experiment 5: Crypt Problem

CSA1717 – CSA1717

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Aim:

To solve the cryptarithmic puzzle **SEND + MORE = MONEY** using Python by assigning digits (0–9) to letters such that the arithmetic equation holds true, ensuring each letter maps to a unique digit and leading digits are non-zero.

Algorithm:

Step 1: Identify Letters

Extract all unique letters from the equation: S, E, N, D, M, O, R, Y

Step 2: Define Constraints

- Each letter must map to a unique digit (0–9).
- Leading letters **S** and **M** cannot be zero.
- The equation must satisfy:

$$SEND + MORE = MONEY$$

Step 3: Generate Permutations

Generate all possible permutations of digits for the 8 letters.

Step 4: Apply Constraints

For each permutation:

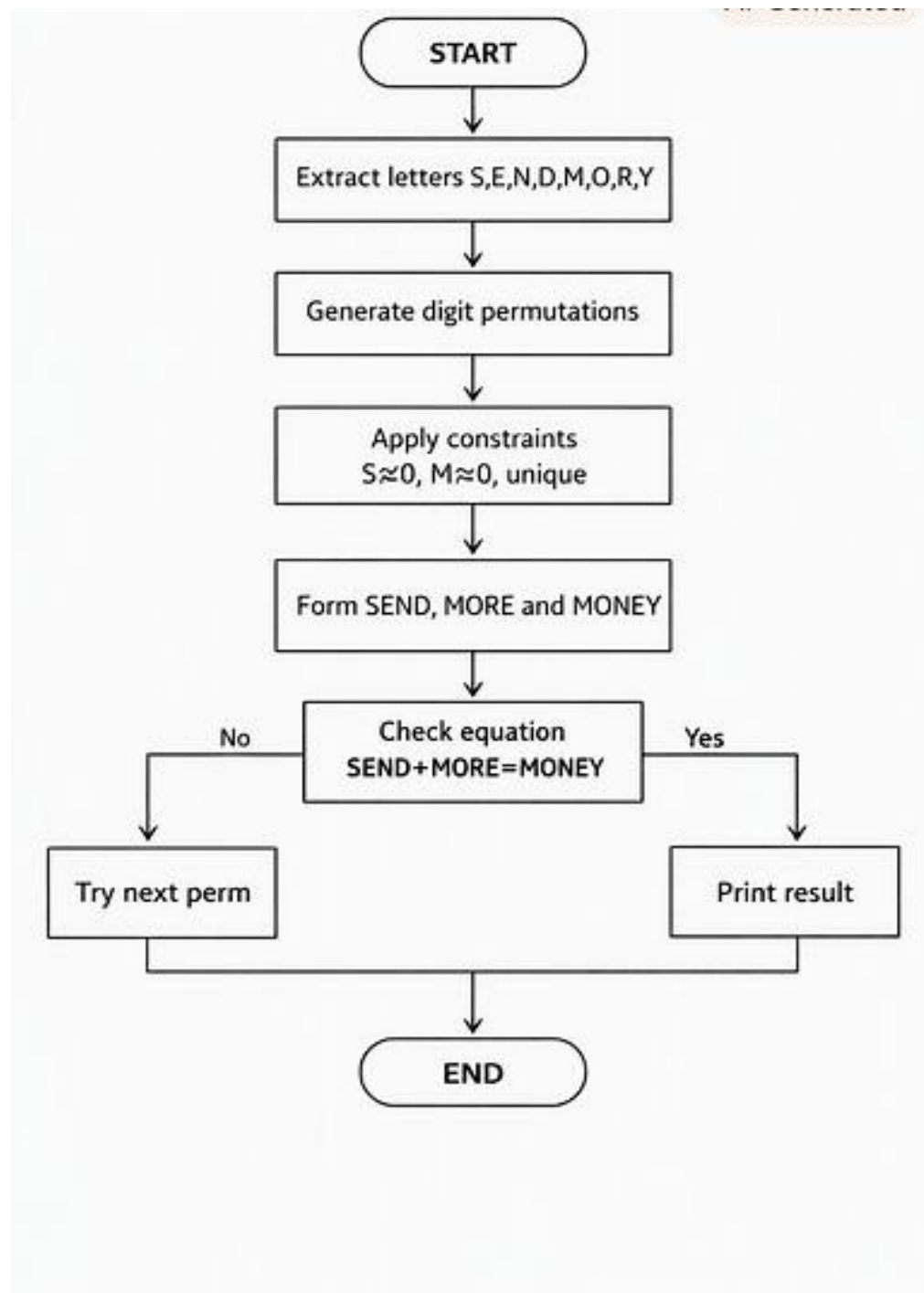
- Check leading digit constraints
- Convert SEND, MORE, MONEY into numbers
- Check if $SEND + MORE = MONEY$

Step 5: Print Solution

If a valid mapping is found, display:

- Letter → Digit mapping
- Numerical values of SEND, MORE, MONEY

Flowchart:



Source Code:

```
# Cryptarithmic Puzzle: SEND + MORE = MONEY
# We solve by assigning digits to letters using permutations.
```

```
import itertools
```

```
def solve_crypt():
```

```
    letters = ('S','E','N','D','M','O','R','Y')
```

```
    digits = range(10)
```

```
    # S and M cannot be zero (leading digits)
```

```
    for perm in itertools.permutations(digits, len(letters)):
```

```
        mapping = dict(zip(letters, perm))
```

```
        if mapping['S'] == 0 or mapping['M'] == 0:
```

```
            continue
```

```
        SEND = (mapping['S']*1000 +
```

```
                mapping['E']*100 +
```

```
                mapping['N']*10 +
```

```
                mapping['D'])
```

```
        MORE = (mapping['M']*1000 +
```

```
                mapping['O']*100 +
```

```
                mapping['R']*10 +
```

```
                mapping['E'])
```

```
MONEY = (mapping['M']*10000 +  
          mapping['O']*1000 +  
          mapping['N']*100 +  
          mapping['E']*10 +  
          mapping['Y'])
```

```
if SEND + MORE == MONEY:  
    print("Solution Found:")  
    print("Mapping:", mapping)  
    print(f"SEND = {SEND}")  
    print(f"MORE = {MORE}")  
    print(f"MONEY = {MONEY}")  
    return
```

```
print("No solution found.")
```

```
# Run the solver
```

```
solve_crypt()
```

Output:

```
===== RESTART: C:/Users/hemal/OneDrive/Documents/SIMATS/AI/Crypt_AI.py =====  
Solution Found:  
Mapping: {'S': 9, 'E': 5, 'N': 6, 'D': 7, 'M': 1, 'O': 0, 'R': 8, 'Y': 2}  
SEND    = 9567  
MORE    = 1085  
MONEY   = 10652
```

Result:

The cryptarithmic puzzle **SEND + MORE = MONEY** was solved by assigning unique digits to each letter such that the arithmetic equation becomes numerically correct.

After applying all constraints and checking valid digit permutations, the correct digit mapping was found as:

- **S = 9**
- **E = 5**
- **N = 6**
- **D = 7**
- **M = 1**
- **O = 0**
- **R = 8**
- **Y = 2**

Using this mapping, the words convert into numbers as follows:

- **SEND = 9567**
- **MORE = 1085**
- **MONEY = 10652**

When adding SEND and MORE:

$$9567 + 1085 = 10652$$

This matches the value of MONEY exactly. Therefore, the solution is **valid**, satisfies all constraints, and correctly solves the cryptarithmic equation.